MIL-D-81514B(AS)
10 September 1969
SUPERSEDING
MIL-D-81514A(AS)
22 April 1968

MILITARY SPECIFICATION

DEVICE, RESTRAINT HARNESS TAKE-UP, INERTIA-LOCKING POWERED-RETRACTING; GENERAL SPECIFICATION FOR

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 The purpose of this specification is to establish the general requirements for all Inertia-Locking, Restraint Harness Take-Up Devices having a powered retraction mode. These devices are intended for use by aircrewmen in designated military aircraft seats during normal flight operations, as well as during and following emergency egress. The detail requirements, if different from those specified herein, shall be covered in the applicable detail specifications. The powered retraction feature of this device is intended to be used for a single operation during the pre-ejection phase of escape to automatically aid the aircrewman in positioning himself properly for ejection.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

NN-P-530	Plywood, Flat Panel
QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard

FSC 1680

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Federal (Continued)

PPP-B-665	Boxes, Paperboard, Metal Stayed, Including Stay Material
PPP-B-676	Boxes, Setup
PPP-C-843	Cushioning Material, Cellulosic

Military

MIL-P-116	Preservation, Methods of
MIL-W-1511	Wire Rope, Steel Carbon Flexible, Preformed
MIL-S-5002	Surface Treatments and Metallic Coatings for Metal Surfaces of Weapon Systems
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5541	Chemicals, Films, and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-F-7179	Finishes and Coatings, General Specification for Protection of Aircraft and Aircraft Parts
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-C-7958	Controls, Push-Pull, Flexible and Rigid
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-F-18264	Finishes: Organic, Weapons System, Application and Control of
MIL-S-18471	Seat System, Ejectable, Aircraft, General Specification for
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices

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Military (Continued)

MIL-A-23121	Aircrew Environmental, Escape and Survival Cockpit Capsule System; General Specification for
MIL-D-23615	Design and Evaluation of Cartridge Actuated Devices
MIL-W-25361	Webbing, Textile, Dacron, Low Elongation
MIL-S-38130	System Safety Engineering of Systems and Associated Subsystems and Equipment; General Specification for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-453	Radiographic Inspection
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	Maintainability Demonstration
MIL-STD-785	Requirements for Reliability Program (for Systems and Equipments)
MIL-STD-831	Test Reports, Preparation of
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MS33586	Metals, Definitions of Dissimilar

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PUBLICATIONS

Department of Defense

MIL-HDBK-132	Protective Finishes

Naval Air Engineering Center

ACEL Report No. 533	Anthropometry of Naval Aviators - 1964 DDC AD 626 322

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government quality assurance representative. Copies of this specification and other unclassified specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Publications and Forms Center (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification - The powered-retracting, inertia-locking, restraint harness take-up device, henceforth referred to herein as PIRD, furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3, 4.3.1 and 6.3). Each device covered by a separate detail specification shall be qualified separately. Before commencing any qualification tests specified herein, Type I releases shall be obtained for the ballistic subsystem and cartridge/gas generator in accordance with MIL-D-23615 and MIL-D-21625, respectively. Before the PIRD may be listed on the applicable qualified products list, Type III or Type IV release, as applicable, shall be obtained for the cartridge in accordance with MIL-D-21625.

3.2 First article - Unless otherwise specified (see 6.2), the PIRD furnished under this specification shall be a product which has been inspected and has passed the first article inspection specified in 4.4 through 4.4.2.

3.3 Selection of specifications and standards - Specifications and standards for necessary commodities and services not specified herein or in the applicable detail specifications shall be selected in accordance with MIL-STD-143.

3.4 Materials - Materials shall conform to applicable specifications and standards and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended. Particular care shall be given to close fitting parts in the choice of both materials and corrosion protection practices.

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3.4.1 Metal parts - All metal parts shall be of the corrosion-resistant type or treated in a manner to render them adequately resistant to corrosion. Unless suitably protected against electrolytic corrosion, dissimilar metals, as defined in MS33586, shall not be used in intimate contact with each other.

3.4.2 Fungus-proof materials - To the greatest extent practicable, the materials used in the device shall not be nutrients for fungi. If materials that are nutrients for fungi must be utilized such materials shall be treated with a fungicidal agent approved by the Naval Air Systems Command.

3.4.3 Corrosion protection - Corrosion protective practices used in the manufacture of these devices shall be in accordance with MIL-F-7179 requirements for exterior surface.

3.4.3.1 Finishes - Protective coatings and finishes used shall not crack, chip or scale during normal service life or in the extremes of atmospheric conditions as specified herein. Surface treatments, coatings and finishes shall conform to MIL-S-5002, or surface treatments herein specified. General guidance in the application and control of organic finishes is provided in MIL-F-18264 and in MIL-HDBK-132.

3.4.3.2 Anodizing - All aluminum and aluminum alloy parts, except those subject to wear, shall be anodized in accordance with MIL-A-8625, Type II. Anodic coatings for all aluminum and aluminum alloy parts subject to wear shall conform to MIL-A-8625, Type III.

3.4.3.2.1 Chemical surface treatment - For aluminum and aluminum alloy parts not subject to wear, abrasion or erosion, chemical conversion surface treatment in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.4.3.3 Plating - Steel parts in contact with aluminum or aluminum alloys shall be cadmium-plated in accordance with QQ-P-416 Type II, Class 1.

3.4.4 Hydraulic fluids - Hydraulic fluids used in the PIRD shall function satisfactorily throughout the temperature range from -65° F to +160° F and shall be nonflammable, noncorrosive, nontoxic and inorganic in nature and shall be approved by the procuring activity.

3.5 Design and construction - The PIRD shall be designed and oriented in such a manner as to be compatible with the designated aircraft seat. It shall be suitable for use with the aircrewman (lower limit 3 percentile man, upper limit 98 percentile man, as specified in the Naval Air Engineering Center (ACEL) Report No. 533), automatically placing him in the best ejection position during the pre-ejection phase of an ejection or in best restrained position under emergency conditions in non-ejection seats. The device shall be designed to meet all of the requirements and tests specified herein.

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3.5.1 Operation - The design of the PIRD shall provide for the following three modes of operation:

(a) Manual lock: Placement of the control handle in the "manual lock" position shall lock the stowage/take-up mechanism precluding additional payout of the personnel restraint harness take-up strap/cable. In addition, as the seat occupant's shoulders move toward the seat back, the stowage/take-up mechanism shall retract the take-up strap/cable as slack is introduced.

(b) Auto-lock: Placement of the control handle in the "auto-lock" position shall unlock the stowage/take-up mechanism permitting the personnel restraint harness take-up strap/cable payout to provide the seat occupant with upper torso mobility, as required. In addition, as slack is introduced in the take-up strap/cable, the stowage/take-up mechanism shall retract the take-up strap/cable. Furthermore, at the prescribed locking "g" level described herein, the stowage/take-up mechanism shall lock, precluding further take-up strap/cable payout.

(c) Powered retraction: Automatic powered retraction of the personnel restraint harness take-up strap/cable, as described herein, shall occur during the pre-ejection sequencing as specified in MIL-S-18471 or MIL-A-23121, as applicable.

3.5.2 Components - The PIRD shall consist of the following components:

(a) Housing

(b) Automatic inertia-locking mechanism

(c) Ballistic subsystem

(d) Restraint harness attachments

(e) Stowage/take-up mechanism

(f). Manual control

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3.5.2.1 Housing - The housing shall integrate satisfactorily with the applicable seat and may be disassembled for service maintenance and repair.

3.5.2.2 Automatic inertia-locking mechanism - An automatic inertia-locking mechanism shall be provided with the performance characteristics required by 3.6.1.1.

3.5.2.3 Ballistic subsystem - The PIRD shall incorporate a ballistic subsystem conforming to MIL-D-23615 with the exceptions specified herein. The CAD or gas generator shall in no way interfere with the normal functioning of the stowage/take-up mechanism.

3.5.2.3.1 Cartridge/gas generator - The cartridge/gas generator shall conform to MIL-D-21625 and may be either contained within or attached to the housing (3.5.2.1) or supplied as a separate unit. Other gas generator/sources, excluding aircraft systems, may be proposed to the procuring activity for approval.

3.5.2.4 Restraint harness attachments - Cable or webbing for restraint harness take-up strap/cable attachment shall conform to MIL-W-1511 or MIL-W-25361, as applicable, and shall be not less than the strength requirements of the 7/32-inch diameter, 7 x 19 construction cable of MIL-W-1511 or the Type II webbing of MIL-W-25361. The cable or webbing shall be securely attached to the stowage/take-up mechanism in such a manner that field replacement can be made readily by one maintenance man without disassembly of the device or removal of the device from the seat. An end fitting shall be attached to the restraint harness take-up strap/cable. The design of the end fitting shall be subject to the approval of the procuring activity. The cable or webbing shall be protected at the point of attachment to prevent damage by snagging or rubbing.

3.5.2.5 Stowage/take-up mechanism - The PIRD shall include a stowage/take-up mechanism for the automatic retraction of the restraint harness in accordance with 3.6.1.3. Unless otherwise specified by the procuring activity or detail specification, when installed in the seat, this mechanism shall permit an 18-inch linear displacement of the personnel restraint attachment hardware as measured from the fully retracted position to the fully extended position of the restraint harness take-up strap/cable.

3.5.2.6 Manual control - The PIRD shall be equipped with an external control for manual operation. The control handle shall have two positions: "auto-lock" and "manual-lock". A push-pull cable meeting the requirements of MIL-C-7958 shall connect the control handle with the locking mechanism. Other control connections may be submitted to the procuring activity for approval for use in lieu of the push-pull cable. End fittings on the control cable shall enable the cable to be disconnected from both the stowage/take-up mechanism and the manual control handle.

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The control handle shall conform to Figure 1. Dimension "A" on Figure 1, shall be specified in the applicable detail seat drawings. For ejection seats, the control handle shall be located as specified in MIL-S-18471. The manual control handle shall include acceptable provisions for its attachment to the seat. Specific envelope and mounting information shall be provided in the applicable detailed seat drawing.

3.5.3 Design load requirements -

3.5.3.1 Proof strength - The PIRD including its mounting provisions, shall withstand a static load of 2,660 pounds. Upon removal of the load, the device shall be capable of meeting the performance requirements of 3.6.

3.5.3.2 Ultimate strength (static) - The PIRD including its mounting provisions, shall withstand a static load of 4,000 pounds without structural failure. Upon removal of the load, the device shall be capable of being manually unlocked in accordance with 3.6.1.2.

3.5.3.3 Crash loads (dynamic) - The system including mounting provisions, as installed in an ejection seat shall be capable of withstanding crash loads as specified in MIL-S-18471 or MIL-A-23121, as applicable. Application of these crash loads shall cause the locking mechanism to lock, as stipulated in 3.6.1.1.1. In addition, the take-up strap/cable even if fully or partially extended shall remain attached to both the device and the personnel restraint hardware under the crash load conditions specified.

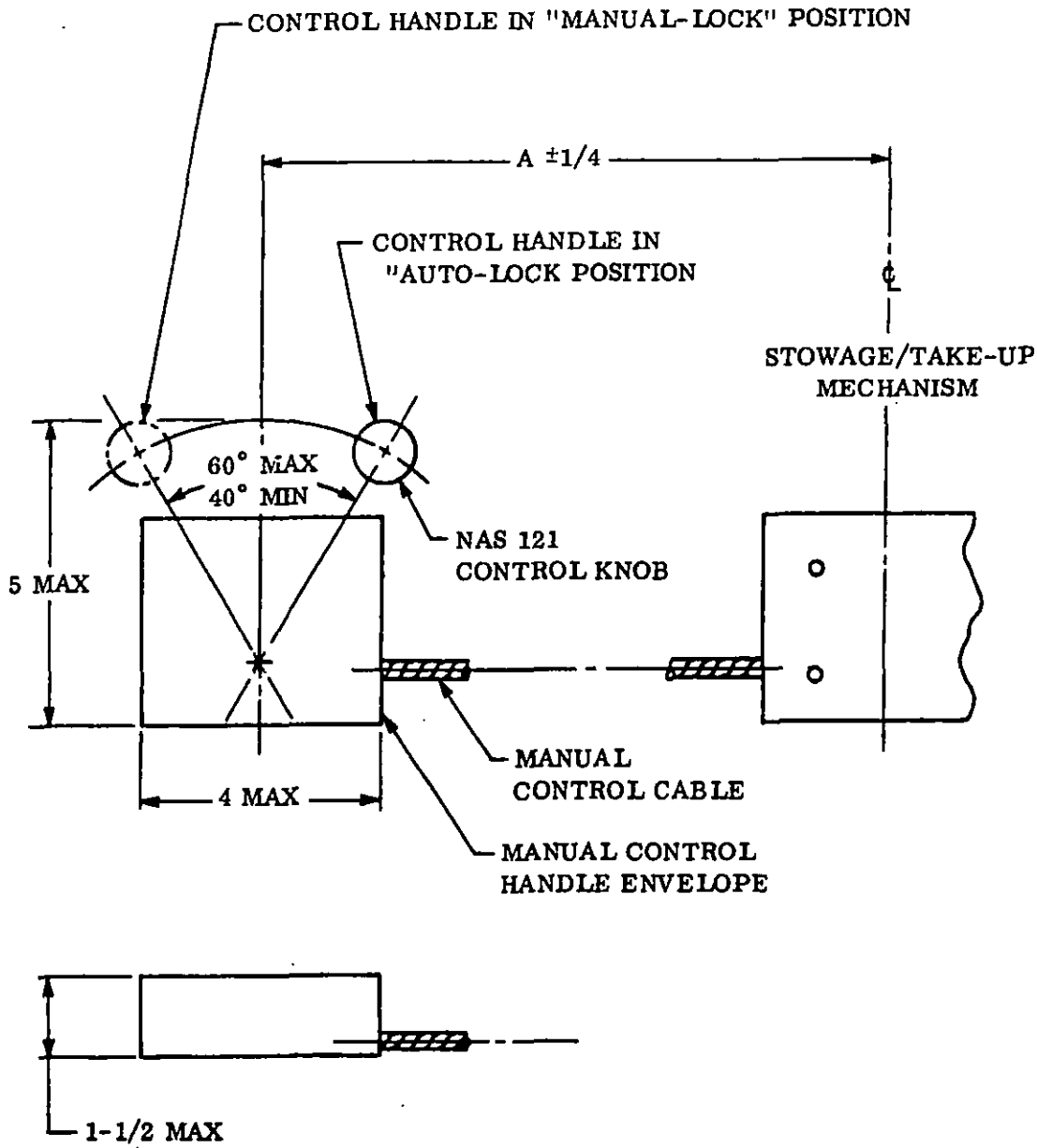
3.6 Performance -

3.6.1 General performance requirements -

3.6.1.1 Inertia-locking mechanism -

3.6.1.1.1 Auto-locking - The following performance requirements shall apply when the crewmember is subjected to forward, longitudinal accelerations, left or right transverse accelerations, or their vectoral sums. With the manual control handle in the "auto-lock" position, the inertia-locking mechanism shall not lock for "g" loads of 2 "g's" or lower. Locking may occur for loads between 2 and 3 "g's". The inertia-locking mechanism shall effectively and positively lock the take-up strap/cable stowage/take-up mechanism when subjected to an inertia load or component thereof of 3 "g's" or more. Once the locking level has been reached, no more than 1/2 inch of take-up strap/cable shall be withdrawn from the stowage/take-up mechanism. The inertia-locking mechanism shall be capable of locking automatically at any extended position of the take-up strap/cable. When the manual control handle is in the "auto-lock" position, and the inertia-lock is engaged, further withdrawal of the restraint harness take-up strap/cable shall not occur. When the inertia load and the

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MANUAL CONTROL CABLE SHOWN DEVELOPED

DIMENSIONS IN INCHES

Figure 1. Manual control dimensions

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load on the take-up strap/cable are relieved, the device shall automatically retract the restraint harness take-up strap/cable. The engaged inertia-lock may be disengaged by a single complete cycling of the manual control handle. The inertia-locking mechanism shall be contained within or shall be attached to the housing.

3.6.1.1.2 Manual locking - The stowage/take-up mechanism shall be locked when the manual control handle is placed in the "manual lock" position.

3.6.1.1.3 Unlocking - When the stowage/take-up mechanism has locked automatically, it shall not unlock unless the manual control handle is cycled from the "auto-lock" position to the "manual lock" position and back to the "auto-lock" position. After manual locking, the stowage/take-up mechanism may be unlocked by moving the manual control handle from the "manual lock" position to the "auto-lock" position.

3.6.1.2 Manual control - For either position, the manual control handle shall remain in the preselected position when subjected to the vibration environment specified in MIL-S-18471 or inadvertent push-pull forces not exceeding 5 pounds. The position of the manual control handle shall be changed by a deliberate movement without requiring other manual operations. When the control handle is mounted on the seat, locking of the stowage/take-up mechanism shall be effected by moving the control handle forward to the "manual lock" position. Unlocking the stowage/take-up shall be effected by moving the control handle rearward to the "auto-lock" position. The stowage/take-up mechanism shall be capable of being unlocked by a maximum load of 25 pounds exerted on the control handle while a load of 50 pounds is applied to the restraint harness take-up strap/cable. The manual control shall lock the stowage/take-up mechanism at any extended position of the take-up strap/cable with a maximum of 1/2-inch additional extension of the cable or webbing.

3.6.1.3 Stowage/take-up mechanism - As installed in the aircraft seat, in the manner intended for service use, it shall retract the restraint harness take-up strap/cable automatically and shall produce a strap/cable tension of not less than 2 pounds or more than 9 pounds throughout the range of extension.

3.6.1.4 Power retraction - As installed and used in the aircraft and in conformance with the applicable seat specification, the powered retracting function of the device, when initiated during the pre-ejection sequence, shall position and forcibly restrain subjects equipped with the applicable personnel and survival equipment. The total residual strap force following full power retraction shall not exceed 100 pounds. The action of the PIRD shall not in itself cause injury to the seat occupant during the powered retraction sequence. During the period while the seat occupant is restrained, he shall not sustain physiological impairments which would degrade his capability to effect the survival and/or evasion phases of his escape. The device, when installed in the seat, shall be capable of performing this function when the seat occupant is subjected to a resisting acceleration of 2 "g's", or 3 "g's" if a seat is not available. Actuation of power retraction shall cause automatic locking of the stowage/take-up mechanism. The locking mechanism shall remain locked unless cycled in accordance with 3.6.1.1.3.

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3.6.1.4.1 Retraction time - The maximum time for retraction of the strap/cable from full extension or any part thereof, from the time that the gas generator firing mechanism is actuated until full retraction is accomplished, shall not exceed 0.3 seconds at ambient or high temperature (70° F and 160° F) for the conditions described herein. Similarly, the maximum time of retraction of the strap at full extension or any part thereof.shall not exceed 0.4 seconds at -65° F.

3.6.1.4.2 Retraction velocity - Within the temperature range and conditioning specified herein, the maximum retraction velocity of the take-up strap/cable shall not exceed 9 ft./sec. after the first 4 inches of strap/cable retraction, for any initial extension of the personnel restraint harness take-up strap/cable. The measurement of strap/cable retraction velocity shall be obtained as close as possible to the inertia reel housing.

3.6.2 Environmental performance requirements - The device shall be capable of performing in accordance with requirements specified herein when subjected to in-service environmental conditions which may include any and all combinations/sequences of the following.

3.6.2.1 Temperature - The system shall withstand and operate within the limits specified herein at temperatures ranging from -65° F to +160° F.

3.6.2.2 Humidity - The system shall withstand and operate within the limits specified herein under the humidity conditions specified in MIL-D-23615.

3.6.2.3 Vibration - The system shall be capable of satisfactory operation after being subjected to the vibration tests specified herein. The system shall operate satisfactorily when subjected to the range of vibrations which may be experienced within the aircraft structural limits during flight, ejection and acceleration.

3.6.2.4 Salt spray - All materials used in the construction of the PIRD shall be corrosion resistant or so processed to withstand the salt spray test of MIL-E-5272, Procedure II.

3.6.2.5 Sand and dust - The system shall be capable of satisfactory operation after exposure to sand and dust in accordance with MIL-E-5272, Procedure I.

3.6.2.6 Altitude - A system shall function properly from sea level to 80,000 feet, mean sea level (0.82 inches of Hg).

3.7 Weight - The maximum weight of the PIRD, including all components listed herein,shall not exceed 5 pounds.

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3.8 Mounting - The PIRD shall be mounted in a suitable manner on the aircraft seat so that it will not vibrate with respect to the seat during service use. The dimensions of the PIRD shall be such as to permit it to fit into the available space allocated for the device in the particular aircraft seat without modification of the existing seat structure. All parts of the device requiring adjustment/inspection by operating squadron personnel shall be readily accessible.

3.9 Special tools - The design of the PIRD shall be such that special tools, fixtures or templates are not required for field replacement of complete device, components and cartridges, or for inspection of the complete assembly.

3.10 Screw threads - Unless otherwise specified, screw threads shall conform to MIL-S-7742. Where practicable, internal or external parts that are threaded shall be positively locked.

3.10.1 Torque requirements of threaded joints - Subassemblies of the PIRD which contain provisions for the attachment of pressure transmission hose or tube assemblies or for the mounting or installation of the device, shall withstand a minimum of 400 inch pounds of breakaway torque. Subassemblies of the device which may be loosened or disassembled by handling, vibration, shipment, etc., shall withstand a minimum of 200 inch pounds of breakaway torque.

3.11 Lubrication - Unless otherwise specified, lubrication shall be in accordance with MIL-STD-838.

3.12 Reliability - Due to the emergency nature of the escape system, prime importance shall be placed upon the attainment of a high overall degree of reliability. A reliability test report should be submitted, indicating the probability of success of the PIRD. The requirement for the probability of success is 0.98 at the 90 percent lower confidence limit. As part of the reliability analysis to be submitted, the contractor shall prepare a failure mode analysis. This analysis shall provide (a) the probability of failure, (b) the expected mode(s), cause(s), of failure, and (c) the consequence of each mode of failure and shall be in accordance with MIL-STD-785.

3.13 Maintainability - The maximum acceptable measure of maintainability for the complete system shall be one man-hour of direct maintenance labor performed at the organizational maintenance echelon per 100 flight hours of the system. The maintainability program to achieve, demonstrate, and assure retention of the system maintainability shall be in accordance with MIL-STD-470 and MIL-STD-471. General design for maintainability guidelines are contained in MIL-STD-1472. Major scheduled maintenance, such as component replacement, shall be designed to occur at Progressive Aircraft Rework periods.

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3.14 Design safety - The contractor shall conform to the general requirements of MIL-S-38130 when designing the PIRD to ensure system design safety. In identifying potential problems/hazards, the contractor shall develop information concerning probability of the problem/hazard occurrence.

3.15 Data requirements - The data requirements of MIL-D-21625 and MIL-D-23615 shall be met. Reports, including copies of applicable films and instrumentation records, shall be furnished in the format and quantities indicated for each test performed in accordance with Section 4, herein, or with MIL-D-21625 and MIL-D-23615. The data and records submitted shall be legible and suitable for analysis to permit the procuring activity and its support activities to conduct an evaluation of the system and its performance capabilities.

3.16 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with respect to installation and performance as specified in MIL-I-8500.

3.17 Identification of product - Identification marking shall be in accordance with MIL-STD-130. Nameplates shall contain the following information:

DEVICE, RESTRAINT HARNESS TAKE-UP, INERTIA-LOCKING, POWERED-RETRACTING
Specification MIL-D-81514B(AS)
Federal Stock No. _____
Manufacturer's part No. _____
Manufacturer's serial No. _____
Contract or Order No. _____
Manufacturer's name or trademark _____
Cartridge nomenclature _____

3.18 Workmanship - The PIRD, including all detail parts, shall be fabricated and finished to produce an item free from defects which could prevent proper functioning of the complete unit in service.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified herein or in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified herein or in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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WARNING

Before commencing any qualification tests specified herein on a new powered-retracting, inertia-locking, restraint harness take-up device; Type I releases shall be obtained for the ballistic subsystem and cartridge/gas generator in accordance with MIL-D-23615 and MIL-D-21625, respectively. Before the PIRD may be listed on the applicable qualified products list, Type III or Type IV release, as applicable, shall be obtained for the cartridge in accordance with MIL-D-21625.

4.2 Classification of inspection - The examination and testing of the powered-retracting, inertia-locking, restraint harness take-up device shall be classified as follows:

(a) Qualification inspection - Qualification inspection consists of the examinations and tests performed on samples submitted for approval as qualified products (see 3.1, 4.3, 4.3.1 and 6.3).

(b) First article inspection - First article inspection consists of examinations and tests performed on PIRD's which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 3.2 and 4.4 through 4.4.2).

(c) Quality conformance inspection - Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification (see 4.5 through 4.5.1.2).

4.3 Qualification inspection - The qualification inspection of the PIRD's shall consist of examinations and those tests summarized in Table I for verifying compliance with all of the requirements of this specification.

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TABLE I

SUMMARY OF TEST SCHEDULE FOR PIRD QUALIFICATION

Group				A	B	C	D	E	F	G
No. of devices				4	3	4	6	4	4	3
Test condition	Test No.	Applicable Specification	Paragraph							
Visual examination	1	81514	4.7.1	4	3	4	6	4	4	3
Radiographic examination	2	81514	4.7.2	4	3	4	6	4	4	3
Threaded joint torque test	3	81514	4.7.12	4	3	4	6	4	4	3
Automatic operation	4	81514	4.7.8	4	3	4	6	4	4	3
Strap/cable tension and retraction	5	81514	4.7.9	4	3	4	6	4	4	3
Manual operation	6	81514	4.7.10	4	3	4	6	4	4	3
Extension and retraction	7	81514	4.7.4.1			1				
Cable and terminal fitting	8	81514	4.7.4.2			1				
Manual control	9	81514	4.7.4.3			1				
Proof strength	10	81514	4.7.3.1			4				
Ultimate strength 1/	11	81514	4.7.3.2			4				
Crash load	12	81514	4.7.3.3			1				
Six foot drop	13	23615	4.5.1.1	1	1	1				
15 "g" shock	14	23615	4.5.1.3	1	1	1	1			
High temperature	15	81514	4.7.5.1	4						

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TABLE I (Continued)

Group				A	B	C	D	E	F	G
No. of devices				4	3	4	6	4	4	3
Test condition	Test No.	Applicable Specification	Paragraph							
Low temperature	16	81514	4.7.5.2		3					
Normal temperature	17	81514	4.7.5.3			4				
Temperature and humidity cycling	18	81514	4.7.5.4				6			
Salt spray	19	81514	4.7.5.5					4		
Sand and dust	20	81514	4.7.5.6						4	
Vibration	21	81514	4.7.5.7							3
Automatic operation	22	81514	4.7.8	4	3	4	6	4	4	3
Strap/cable tension and retraction	23	81514	4.7.9	4	3	4	6	4	4	3
Manual operation	24	81514	4.7.10	4	3	4	6	4	4	3
Lock shut test	25	23615	4.2						1	1
Powered retraction functional Phase I, *Cond.* (a)	26	81514	4.7.7.1				1	1		
(b)	27	81514	4.7.7.1				1			
(c)	28	81514	4.7.7.1				1	1		
(d)	29	81514	4.7.7.1	1	1		1	1		

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TABLE I (Continued)

Group				A	B	C	D	E	F	G
No. of devices				4	3	4	6	4	4	3
Test condition	Test No.	Applicable Specification	Paragraph							
(e)	30	81514	4.7.7.1	1	1		1		1	
(f)	31	81514	4.7.7.1	1	1		1			1
(g)	32	81514	4.7.7.1	1						
Temperature - Altitude Test	33	81514	4.7.7.1.1						2	
Powered retraction functional Phase II, Cond. (a)	34	81514	4.7.7.2			1				
(b)	35	81514	4.7.7.2			1				
(c)	36	81514	4.7.7.2							1
(d)	37	81514	4.7,7.2			1				
(e)	38	81514	4.7.7.2					1		
(f)	39	81514	4.7.7.2			1				

1/ To be conducted with a device which has undergone Ballistic Firing.

4.3.1 Qualification samples - Unless otherwise specified in the applicable detail specification, the qualification samples shall consist of twenty-eight PIRD's and cartridges of each manufacturer's part number for which qualification is desired. The units shall comply in all respects to the specification requirements herein and in the applicable detail specification. Samples shall be forwarded to the Supply Officer, Naval Air Development Center, Johnsville, Warminster, Pennsylvania 18974, Attention: Director, Aerospace Crew Equipment Department.

4.3.1.1 Each sample shall be plainly identified by a securely attached durable tag marked with the following information:

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Samples for qualification inspection
DEVICE, RESTRAINT HARNESS TAKE-UP,
INERTIA-LOCKING, POWERED-RETRACTING
Manufacturer's designation or number
Name of manufacturer
Submitted by (name) (date) for qualification
inspection in accordance with the requirements
of MIL-D-81514B(AS) under authorization
(reference authorizing letter).

4.4 First article inspection - The first article inspection of the PIRD's shall consist of the examinations and tests listed in Table II.

TABLE II

FIRST ARTICLE INSPECTION OF PIRD

GROUP	A	B
NO. OF DEVICES	4	1
Visual examination	4	1
Radiographic examination	4	1
Automatic operation	4	1
Strap/cable tension and retraction	4	1
Manual operation	4	1
Proof strength	4	1
Weight	4	1
High temperature		1
Low temperature		1
Salt spray		1
Sand and dust		1
Powered retraction mode functional tests:		
3 percentile fully extended 160° F	1	
98 percentile fully extended 70° F	1	1
3 percentile 1/2 extension 70° F	1	
98 percentile 1/4 extension -65° F	1	

4.4.1 First article samples - Unless otherwise specified, as soon as practicable after award of the contract or order, the manufacturer shall submit five PIRD's and cartridges as first article samples. Unless otherwise directed by the contracting officer, the first article inspection samples shall be furnished to the Government for performance of the first article inspection. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these devices

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from contract to contract, submission of further first article samples may be waived at the discretion of the procuring activity (see 6.2 (b)). Approval of the first article samples or the waiving of first article inspection does not preclude the requirements of submitting to the quality conformance inspection as directed by the contracting officer (see 6.2 (c)). The samples shall be plainly identified by securely attached durable tags marked with the following information:

> Samples submitted by (name) (date) for first article inspection in accordance with the requirements of MIL-D-81514B(AS) under Contract No. _____.

4.4.2 Upon completion of the first article inspection, all the applicable inspection reports and when applicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the cognizant Government activity. One, used, approved PIRD shall be returned to the manufacturer for use in monitoring production and shall not be considered as part of the quantity to be delivered under contract. The other devices in the sample shall be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract.

4.5 Quality conformance inspection - The sampling and inspection levels shall conform to MIL-STD-105. The quality conformance inspection shall consist of the following:

> Visual examination (see classification of defects)
> Radiographic examination
> Automatic operation
> Strap/cable tension and retraction
> Manual operation
> Proof strength
> Weight
> Preparation for delivery

4.5.1 Sampling -

4.5.1.1 Inspection lot -

4.5.1.1.1 Powered-retracting, inertia-locking, restraint harness take-up device - An inspection lot size shall be expressed in units of one PIRD, complete with all components. An inspection lot shall consist of PIRD's complete with all components. The sample unit shall be one PIRD.

4.5.1.1.2 Preparation for delivery - An inspection lot shall be expressed in units of one fully prepared shipping container, containing PIRD's. An inspection lot shall consist of shipping containers fully prepared for delivery from essentially the same materials and components. The sample unit shall be one shipping container, containing PIRD's fully prepared for delivery with the exception that it need not be sealed.

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4.5.1.2 Sampling for tests and examinations of the PIRD - The sample size, acceptance criteria, test and examinations required for the PIRD's shall be as specified in Table III.

TABLE III

QUALITY CONFORMANCE INSPECTION
SAMPLE SIZE, ACCEPTANCE CRITERIA, TESTS, AND EXAMINATIONS
OF THE POWERED-RETRACTING, INERTIA-LOCKING, RESTRAINT
HARNESS TAKE-UP DEVICE

Inspection	Method paragraph	Sample size	Acceptance criteria 1/
Visual examination	4.7.1	Every device for critical defects. Level II for minor defects 1/	Reject all units with any critical defects and an acceptable quality level of 2.5 defects per hundred units for minor defects.
Radiographic examination	4.7.2	Every device	Reject all defective units.
Automatic operation	4.7.8	Every device	Reject all defective units.
Strap/cable tension and retraction	4.7.9	Every device	Reject all defective units.
Manual operation	4.7.10	Every device	Reject all defective units.
Proof strength	4.7.3.1	Inspection Level II 1/	An acceptable quality level of 1.5 defects per 100 units.
Weight	4.7.11	Inspection Level II 1/	An acceptable quality level of 1.5 defects per 100 units.
Preparation for delivery	4.7.1.2	Inspection Level S-2 1/	An acceptable quality level of 4.0 defects per 100 units.

1/ The sampling plan acceptance numbers shall apply collectively to all the characteristics within a stated acceptable quality level.

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4.6 Test conditions and procedures -

4.6.1 Atmospheric conditions - Unless otherwise specified tests shall be conducted at local ambient temperatures and barometric pressure, and corrections shall be made to provide agreement with the temperature and pressure calibration of the instruments. Test data provided by any instrument not calibrated to Standard Temperature Pressure (STP) conditions shall be corrected to determine STP requirements. STP conditions are 29.92 inches of mercury and 70 degrees Fahrenheit.

4.6.2 Test equipment - Environmental test equipment shall be in conformance with the requirements of MIL-E-5272 and shall be capable of producing and maintaining the required environmental conditions. All other test equipments shall be capable of producing and maintaining the test conditions herein specified, in a manner acceptable to the procuring activity. The facilities of the Aerospace Crew Equipment Department may be utilized in performing some of the required tests in order to preclude the purchasing of costly duplicate test equipment. The Aerospace Crew Equipment Department has an ejection tower which is capable of producing ejection load type phenomena and a linear accelerator which is capable of producing crash load phenomena and load conditions required by Table VI.

4.6.2.1 Instrumentation - Any state-of-the-art type instrumentation and recording equipment which conform to acceptable laboratory standards and which are appropriate for measuring the conditions and operating characteristics concerned may be used. However, the accuracy of all test apparatus shall be calibrated periodically to the satisfaction of the procuring activity and documented. Documented calibration records shall be maintained by the manufacturer for a period not less than five years following delivery and shall be available for inspection by the cognizant activity or procuring activity.

4.6.3 Test coverage and reporting - Performance parameters such as the following shall be measured simultaneously and recorded during ballistic tests of the PIRD:

(a) Time of gas generator initiation.

(b) Take-up strap/cable force-time for the entire retraction sequence (including residual take-up strap/cable force after retraction).

(c) Retraction rate of the take-up strap/cable into the housing for the entire retraction sequence.

(d) Opposing load for conditions (a), (b) and (c) of Table VI.

(e) Pressure-time history of the cartridge/gas generator during test.

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Two independent means shall be provided to demonstrate the retraction time and velocity of the PIRD when subjected to the applicable tests specified herein. Other data, such as strength and environmental conditioning data, shall be measured and recorded as necessary. A test report conforming to the requirements of MIL-STD-831 and containing the data matrix developed during the qualification tests shall be submitted to the cognizant Government activity. Test reports also shall be prepared by the testing activity for all tests performed as part of first article inspection (4.4) or as part of quality conformance inspection (4.5) and shall be submitted to the cognizant and procuring Government activities.

4.7 Inspection methods -

4.7.1 Visual examination -

4.7.1.1 PIRD - Every PIRD shall be examined visually to determine conformance to all the requirements of this specification and applicable drawings with respect to material, workmanship, design, standard parts, weight, finish dimensions and marking. The classification of defects, Table IV, shall be used to classify the defects.

TABLE IV

CLASSIFICATION OF DEFECTS FOR VISUAL EXAMINATION OF THE POWERED-RETRACTING, INERTIA-LOCKING, RESTRAINT HARNESS TAKE-UP DEVICE.

Critical	Minor
1. Dimensions not in conformance with applicable drawings or within specified tolerances.	201. Marking-missing, insufficient, incorrect, illegible or not permanent.
2. Device does not fit into the applicable seat due to interference.	
3. Material imperfections, foreign matter imbedded, burrs.	
4. Any component missing, malformed, fractured or otherwise damaged.	
5. Any frayed or cut cable or webbing.	
6. Any component loose or otherwise not securely retained.	

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TABLE IV (Continued)

Critical	Minor
7. Incorrect assembling or improper position of components. 8. Any functioning part that works with difficulty. 9. Faulty workmanship or other irregularities.	

4.7.1.2 Preparation for delivery - Each of the fully prepared shipping containers, containing PIRD's of one type, selected as a sample unit from the lot shall be examined to determine that the packaging, packing, and marking conform to this specification. The classification of defects, Table V, shall be used to classify the defects found.

TABLE V

CLASSIFICATION OF DEFECTS FOR PREPARATION FOR DELIVERY

Item	Major defects
Exterior and Interior Markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence of method of application; markings not the same on the interior and exterior containers.
Packaging and Packing Materials	Any nonconforming component; any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, intermediate package, container flaps, or loose strappings, etc.; bulging or distortion of the containers.
Exterior and Interior Weight or Content	Number per container is more or less than required; gross or net weight exceeds the requirements.

4.7.2 Radiographic examination - Every PIRD shall be examined radiographically in accordance with MIL-STD-453 for detection of internal defects and confirmation of the presence and proper assembly of parts.

4.7.3 Strength tests -

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4.7.3.1 Proof strength - The PIRD shall be mounted to simulate its installation on the seat. The manual control handle shall be placed in the "manual lock" position and a static load of 2,660 pounds shall be applied to the take-up strap/cable fitting with a spring dynamometer or standard testing machine. This test shall be conducted with the take-up strap/cable in the following extensions: fully retracted, half extension and fully extended. Upon removal of the load the stowage/take-up mechanism shall operate freely, allowing the take-up strap/cable to pay out when the control handle is moved to the "auto-lock" position. Assemblies subjected to the proof strength test shall then be subjected to and meet the requirements specified in 4.7.8, 4.7.9 and 4.7.10. In accordance with requirements of 3.5.3.1, there shall be no malfunction or failure of the device or its mounting provisions.

4.7.3.2 Ultimate strength - The PIRD shall be mounted to simulate its installation on the seat. The manual control handle shall be placed in the "manual lock" position and a static load of 4,000 pounds shall be applied to the take-up strap/cable fitting with a spring dynamometer or a standard testing machine. This test shall be conducted with the take-up strap/cable fully retracted and extended half-way. In accordance with the requirements of 3.5.3.2, the device and its mounting provisions shall not structurally fail, and upon removal of the load, the stowage/take-up mechanism shall function normally.

4.7.3.3 Crash loads (dynamic) - The PIRD shall be mounted on the ejection seat and the assembly shall be tested in accordance with MIL-S-18471. In accordance with the requirements of 3.5.3.3, the device and its mounting provisions shall not fail. The device need not function following the test.

4.7.4 Life - One sample device shall be subjected to each of the following tests:

4.7.4.1 Extension and retraction - The device shall be subjected to a life test consisting of 5,000 cycles of shoulder harness extension and retraction. Each cycle shall consist of fully extending the cable or webbing and allowing the cable or webbing to return to the fully retracted position. At least 1,000 cycles shall be completed with the axis of the cable or webbing extended at an angle of 30 degrees with the axis of the cable or webbing at the point of entry into the device housing. After each 10 cycles, the cable or webbing shall be fully extended, the control handle placed in the "manual lock" position, and the cable or webbing allowed to retract. Before the test is begun, twice during the test, and following completion of the test, the cable or webbing shall be fully extended, the control handle placed in the "manual lock" position, and the maximum cable or webbing movement measured at four (4) different extended positions (no less than 1/8 of full extension apart) to determine that the movement complies with the requirement of 3.6.1.2 and does not exceed 1/2-inch. Subsequent to completion of the life test, the device shall demonstrate compliance with 4.7.8, 4.7.9 and 4.7.10. There shall be no excessive wear on the external or

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internal surface of the device mechanism. The device shall then be subjected to the proof strength specified in 4.7.3.1.

4.7.4.2 Cable and terminal fitting - The device shall be tested as follows: While in the fully retracted position the cable shall be bent until the axis of the cable terminal is inclined at an angle of 60 degrees with the axis of the cable at the point of entry into the device housing. Maintaining this angle, the cable terminal shall be rotated, generating a conical surface, about the axis of the cable at the point where the cable enters the reel housing. Each 360 degrees of rotation shall constitute 1 cycle. The fitting shall be subjected to 2,500 cycles with the control handle in the "auto-lock" position, and to 2,500 cycles with the control handle in the "manual lock" position. After each 10 cycles the cable shall be fully extended and permitted to return to the retracted position. Upon completion of this test, the device shall be subjected to the proof strength test specified in 4.7.3.1. The cable and terminal fitting shall then be inspected, and there shall be no breakage or excessive wear on either the cable or the terminal fitting.

4.7.4.3 Manual control - With the device and manual control installed on the seat or on a simulated mounting, the manual control mechanism shall be subjected to 5,000 cycles of locking and unlocking. Each cycle shall consist of moving the manual control handle from the locked position to the unlocked position and then back to the locked position. Upon completion of this test, the device shall be tested as specified in 4.7.8, 4.7.9 and 4.7.10. There shall be no mechanism failure.

4.7.5 Environmental tests - The environmental tests required and performed as part of first article inspection (4.4) shall be performed in accordance with 4.7.5.1, 4.7.5.2, 4.7.5.5 and 4.7.5.6, except that the device shall not be fired until it has been subjected to all four conditions. Environmental tests conducted as part of qualification inspection (4.3) involve ballistic firing as noted. Devices shall be subjected to each of the following environmental tests, in the order given:

4.7.5.1 High temperature - The device(s) shall be conditioned at a temperature of +160° F for a period one hour longer than that required to reach thermal equilibrium. While at this temperature, the devices shall be tested as specified in 4.7.8, 4.7.9 and 4.7.10. There shall be no malfunction or failure of the devices. They shall then be ballistically fired in accordance with Table I. The powered retraction performance shall comply with the requirements of 3.6.1.4.

4.7.5.2 Low temperature - The device(s) shall be conditioned at a temperature of -65° F for a period one hour longer than that required to reach equilibrium. While at this temperature, the devices shall be tested as specified in 4.7.8, 4.7.9 and 4.7.10. There shall be no malfunction or failure of the devices. They shall then be ballistically fired in accordance with Table I. The powered retraction performance shall comply with the requirements of 3.6.1.4.

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4.7.5.3 Normal temperature - The device(s) shall be conditioned at a temperature of +70 ±5° F. While at this temperature they then shall be ballistically fired in accordance with Table I. They shall comply with the performance requirements of 3.6.1.4.

4.7.5.4 Temperature and humidity cycling - Temperature and humidity cycling tests shall be in accordance with the test schedule of MIL-D-23615, 4.5.2, with the following exceptions:

(a) Six devices shall be used for this test.

(b) Low temperature shall be -65° F in lieu of -80° F.

(c) Two devices shall be removed at each of the prescribed intervals and tested as specified in 4.7.8, 4.7.9 and 4.7.10. There shall be no malfunction or failure. They shall then be fired in accordance with Table I. The powered retraction performance shall comply with the requirements of 3.6.1.4.

4.7.5.5 Salt spray - The device(s) shall be tested in accordance with the salt spray Procedure II of MIL-E-5272. At the end of the test period and 48 hours later, the devices shall be subjected to the tests specified in 4.7.8, 4.7.9 and 4.7.10. There shall be no malfunction or failure of the device(s). They shall be ballistically fired in accordance with Table I. The powered retraction performance shall comply with the requirements of 3.6.1.4.

4.7.5.6 Sand and dust - The device(s) shall be tested in accordance with Procedure I for sand and dust of MIL-E-5272. After exposure to the test, the device(s) shall be subjected to the tests specified in 4.7.8, 4.7.9 and 4.7.10. There shall be no malfunction or failure of the devices. They shall then be ballistically fired in accordance with Table I. The powered retraction performance shall comply with the requirements of 3.6.1.4 or 3.3.5.1 of MIL-D-23615, as applicable.

4.7.5.7 Vibration tests - Test unit(s) shall be exposed along each of three mutually perpendicular axes to the vibration of the frequency and amplitude specified below. The units shall be secured to a rigid fixture capable of transmitting the vibration conditions. The units shall be tested in both the "manual lock" and in the "auto-lock" position during the inflight cycling test and in the "manual lock" position during the ejection cycling test, with tension maintained in the strap to hold it extended approximately six (6) inches. Following conclusion of the flight resonance tests, the flight cycling test (see 4.7.5.7.2.1) shall be divided into 3 parts, the first 15 minutes being conducted at -65° F, the second part at normal room temperature, and the final 15 minutes at +160° F. When in the "manual lock" position, the device shall not unlock when subjected to the required vibration spectra. Nor shall the device lock when in the "auto-lock" position and when subjected to the required

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vibration spectra. After completion of the vibration tests, the device shall be visually examined for any defects and shall be subjected to the tests specified in 4.7.8, 4.7.9 and 4.7.10. There shall be no malfunction or failure of the devices. They shall then be ballistically fired in accordance with Table I. The powered retraction performance shall comply with the requirements of 3.6.1.4 or 3.3.5.1 of MIL-D-23615, as applicable.

(a) Vibration spectra for flight conditions

5-10 cps	0.080 inch double amplitude
10-28 cps	±0.41g
28-65 cps	0.010 inch double amplitude
65-500 cps	±2.0g

(b) Vibration spectra for ejection conditions

5-10 cps	0.20 inch double amplitude
10-18 cps	±1.0g
18-80 cps	0.06 inch double amplitude
80-2,000 cps	±20.0g

4.7.5.7.1 Resonance tests - The following resonance tests shall be performed:

4.7.5.7.1.1 Flight resonance tests - The resonance modes of the unit shall be determined by varying the frequency of applied vibration slowly between 5 and 500 cps at acceleration levels specified in 4.7.5.7 (a). Individual resonance surveys shall be conducted with vibration applied along each axis of the set of three mutually perpendicular axes. If more than four resonances are encountered with vibration applied along any one axis, the four most severe resonances shall be chosen for test. Vibrate at each resonant condition for a period of 30 minutes at the applicable amplitude specified in 4.7.5.7(a) at 70° F only.

4.7.5.7.1.2 Ejection resonance tests - The resonant modes of the unit shall be determined by varying the frequency between 5 and 2,000 cps at acceleration levels specified in 4.7.5.7(b). If more than four resonances are encountered with vibration applied along one axis, the four most severe resonances shall be chosen for test. Vibration at each resonant frequency shall be maintained at the indicated resonant condition for a period of 3 minutes at the applicable amplitude specified in 4.7.5.7(b) and at 70° F.

4.7.5.7.2 Cycling tests - The following cycling tests shall be conducted:

4.7.5.7.2.1 Flight cycling test - The unit shall be vibrated with the frequency varying between 5 and 500 cps at the amplitude specified in 4.7.5.7(a). The rate of frequency change shall be logarithmic. When there is no provision for logarithmic cycling, a linear rate of frequency change may be used. The cycling time shall be such that 15 minutes are required to vary from 5 to 500 and back to 5 cps. Before

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the final complete cycle, the manual control shall be placed from "auto-lock" into "manual lock" position. The time duration for each axis shall be as follows:

No resonant frequency	3.0 hours
1 resonant frequency	2.5 hours
2 resonant frequencies	2.0 hours
3 resonant frequencies	1.5 hours
4 resonant frequencies	1.0 hours

4.7.5.7.2.2 Ejection cycling test - The unit shall be vibrated with the frequency varying between 5 and 2,000 cps at the amplitudes specified in 4.7.5.7(b). The rate of frequency change shall be logarithmic unless provisions for such cycling are unavailable and then a linear variation may be used. The cycling time shall be such that 15 minutes are required to vary from 5 to 2,000 and back to 5 cps. The test shall consist of one sweep each at -65° F,+70° F and +160° F for each axis.

4.7.6 Ballistic subsystem - The ballistic subsystem device of the PIRD shall be tested in accordance with MIL-D-23615, for Type I release as specified in 3.1, all tests shall be conducted in accordance with the tests listed in Table I.

4.7.6.1 Cartridge - The cartridge shall be in accordance with requirements of MIL-D-21625 for applicable Type III or Type IV release, as applicable. Cartridges used for the cartridge tests and for the PIRD qualification inspection under 4.3 shall be from the same lot.

4.7.7 Powered retraction mode - The PIRD shall complete the following powered retraction mode tests using all components of the device as a complete assembly. Testing shall be performed in two phases. Before commencing Phase II (4.7.7.2) of this test program, the device shall have completed successfully the Phase I (4.7.7.1) tests specified in Table VI and 4.7.7.1.1 herein and shall have obtained release from the procuring activity for testing of the device using human subjects. It is preferred that the device be tested while mounted in the seat for which intended. If a seat is not available, these mounting provisions shall be simulated in a manner acceptable to the testing activity and the resisting acceleration, where required, shall be increased to 3 "g's".

4.7.7.1 Powered retraction mode functional tests, Phase I - A 3 or 98th percentile anthropomorphic dummy, as noted in Table VI, dressed in full flight clothing and survival equipment in a manner acceptable to the testing activity shall be installed in the seat with all personnel restraints connected. If the intended seat is not available then the device shall be tested on a test rig that simulates the unit's installation and interfaces with the intended seat. The PIRD shall then be fired unde the conditions described in Table VI to determine compliance with performance requirements specified herein. For each of the conditions described in Table VI, the following performance parameters shall be measured simultaneously and recorded:

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1. Time of gas generator initiation.
2. Take-up strap/cable force - time for the entire retraction sequence (including residual take-up strap/cable force after retraction).
3. Retraction rate of the take-up strap/cable into the housing for the entire retraction sequence.
4. Opposing load for conditions (a), (b) and (c) Table VI.
5. Pressure-time history of the cartridge/gas generator during above tests.

The performance of the PIRD shall comply with the requirements of 3.6.1.4, 3.6.1.4.1 and 3.6.1.4.2.

4.7.7.1.1 Temperature-altitude test - The PIRD shall be conditioned at a temperature of -65° F until equilibrium is reached. The absolute pressure in the chamber shall correspond to an altitude of 80,000 feet above mean sea level (0.82 inches of Hg). When these conditions are attained the device shall be functionally tested as follows:

(a) Prior to conditioning, the device shall be mounted on a suitable test rig. Hanging from the fully extended strap shall be a 75-pound weight. After reaching equilibrium, the device shall be actuated and shall vertically lift the weight to a fully retracted position. The device shall remain intact. Retraction time and strap/cable velocity shall be recorded for information purposes only.

(b) A second device shall be subjected to the same environment and then shall be brought back to an equilibrium condition of sea level altitude pressure and +70° F. It shall be tested in the same manner as described above and shall fully retract the weight. The device shall remain intact.

4.7.7.2 Powered-retraction mode functional tests, Phase II - Following the successful completion of the tests required under 4.7.7.1 and 4.7.7.1.1 and procuring agency release of the PIRD for testing with human subjects, the following test program of Table VII shall be conducted.

4.7.8 Automatic operation - Each device shall be subjected three times to the auto-lock test specified below.

4.7.8.1 Methods - Means shall be provided for extending and retracting the restraint harness take-up strap/cable and obtaining an accurate calibration to indicate that the "g" load at which the device locks is within the tolerances specified in 3.6.1.1.1. This test shall be accomplished by starting with the device in a static condition with an increase in "g's" from zero to 2 "g's" under which condition the device shall not lock, and an increase in load from zero to 3 "g's" under which condition the device shall lock. An approved tester equivalent to Pacific Scientific Inertia Reel tester No. 3101101-1 shall be used for this test. A specification control drawing of this tester is available through the procuring activity.

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TABLE VI

POWERED RETRACTION MODE FUNCTIONAL TESTS

Load condition	Dummy position	Temperature condition	Dummy percentile
(a) An opposing load caused by a 2 "g" deceleration	fully extended forward	+70° F	3 and 98
(b) An opposing load caused by a 2 "g" deceleration	fully extended and sideward to the maximum extent permitted by the geometry of the seat and cockpit. The accepting activity shall determine the extent of sideward displacement	+70° F	98
(c) An opposing load caused by a 2 "g" deceleration	1/4 of full extension	+70° F	3 and 98
(d) Normal load. No forward/sideward acceleration/deceleration	fully extended	+160° F + 70° F - 65° F	3 3 and 98 98
(e) Normal load. No forward/sideward acceleration/deceleration	1/2 of full extension	+160° F + 70° F - 65° F	98 3 and 98 3
(f) Normal load. No forward/sideward acceleration/deceleration	1/4 of full extension	+160° F + 70° F - 65° F	3 3 and 98 98
(g) Normal load. No forward/sideward acceleration/deceleration	with dummy position full back against the the seat back or equipment mounted on seat back	+160° F	98

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TABLE VII 1/

Load condition	Subject position	Subject percentile
(a) 2 g	full out	98
(b) none	full out	98
(c) none	full out	3
(d) none	1/2	98
(e) none	1/4	98
(f) none	1/4	3

1/ All tests at 70° F.

4.7.8.2 Auto-lock test - The PIRD may be tested in one position only, if the orientation of the device definitely will not affect the locking point. If the orientation of the device might affect the locking point, the device shall be tested in five positions. These positions shall be such that the acceleration imposed upon the device by the testing machine simulates rearward, 45 and 90 degrees right transverse and 45 and 90 degrees left transverse acceleration in a horizontal plane. With the device in each of these five positions, or in one convenient position if the orientation will not affect the locking point, and with the control handle in the "auto-lock" position and the device fixed in position, the take-up strap/cable shall be subjected to an increasing "g" load from zero to a value of 2 "g's". The 2 "g's" shall be maintained for at least 1 inch of strap/cable travel. The device shall not lock under this condition. The device then shall be subjected to an increasing "g" load from zero to 3 "g's", with 3 "g's" maintained for at least 1 inch of travel or until the stowage/take-up mechanism locks. Both, the time locking occurs and the input acceleration level, shall be recorded. Once the locking level has been reached, no more than 1/2 inch of take-up strap/cable shall be withdrawn from the stowage/take-up mechanism. To verify that the preceding condition has been met, a measurement of the take-up strap/cable extension before and after locking, exclusive of the travel necessary to reach the "g" locking level and the take-up strap/cable stretch or packing, shall be recorded. The "auto-lock" test shall be conducted with the take-up strap/cable at 1/4, 1/2 and 3/4 of full extension.

4.7.9 Take-up strap/cable tension and retraction - With the control handle in the "auto-lock" position, the tension in the take-up strap/cable shall be measured with a spring scale. Readings shall be taken at 2-inch intervals until the strap/cable is fully extended. Strap/cable tension shall be in accordance with the values specified in 3.6.1.3. With the take-up strap/cable fully extended, the stowage/take-up mechanism shall fully retract the take-up strap/cable with the manual control handle in either the "manual lock" or "auto-lock" position.

4.7.10 Manual operation - With the device and manual control assembly simulating mounting as shown on Figure 1, the control handle shall be moved from the "auto-lock" to the "manual lock" position. This movement shall be within the

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limits specified in 3.5.2.6, and it shall cause the stowage/take-up mechanism to lock positively and completely. A 15-pound load shall be applied to the take-up strap/cable and the amount of cable or webbing which is payed out shall not exceed 1/2 inch. The device shall not become unlocked unless the control handle is placed in the "auto-lock" position by the operator.

4.7.11 Weight - The weight of the device shall be checked for conformance with the applicable drawings and shall not exceed 5 pounds in conformance with 3.7.

4.7.12 Threaded joint torque test - The PIRD shall be tested to determine that the torque requirements of 3.10.1 are met.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - Unless otherwise specified, each assembly shall be preserved in accordance with Method 1A of MIL-P-116 and packaged in a snug fitting container conforming to PPP-B-636, or PPP-B-665, or PPP-B-676. Sufficient cushioning material conforming to PPP-C-843 shall be used to prevent movement of the preserved assembly within the unit container.

5.1.1 Intermediate containers - Packaged assemblies shall be over-packaged in snug-fitting intermediate containers conforming to PPP-B-636. Intermediate containers shall contain not more than 10 unit packages. As near as possible, intermediate containers shall contain the same number of unit packages.

5.2 Packing - Packing shall be Level A, B, or C, as specified (see 6.2).

5.2.1 Level A - Assemblies preserved and packaged as specified in 5.1 shall be packed in containers conforming to overseas type of PPP-B-621, or Style A or B of PPP-B-601, using plywood conforming to NN-P-530. Containers shall be closed and strapped in conformance with the appendix of the applicable specification. Weight of packed containers shall not exceed 200 pounds.

5.2.2 Level B - Assemblies preserved and packaged as specified in 5.1 shall be packed in fiberboard boxes conforming to Type II, Class 2 of PPP-B-636. Boxes shall be closed and strapped in accordance with the appendix of the box specification.

5.2.3 Level C - Assemblies preserved and packaged as specified in 5.1 shall be packed in a manner to assure acceptance by common carrier for transportation at the lowest applicable rate, and safe delivery to the first receiving activity. Containers shall conform to Uniform Classification Rules or to other regulations applicable to the mode of transportation to be used.

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5.3 Marking - In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use - The PIRD is intended for use in designated Military aircraft seats during normal flight operations as well as during and following emergency egress and during crash conditions. The inertia-locking mechanism is intended as a safety device for use by seat occupants to restrain movement during a crash or deceleration of the aircraft. The powered retraction feature of this device is intended to be used for a single operation during the pre-ejection phase of escape to automatically aid the aircrewman in positioning himself properly for ejection.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification or applicable detail specification.

(b) Whether first article inspection is required (see 4.4.1).

(c) Name and address of the first article inspection laboratory (see 4.4.1).

(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(e) Items of data required (see 6.4).

6.3 Qualification - With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command, Department of the Navy, Washington, D. C. 20360; however, authorization for qualification of products shall be obtained from the Commanding Officer, Naval Air Development Center, Johnsville, Warminster, Pennsylvania 18974, Attention: Director, Aerospace Crew Equipment Department. Prior to submission of the samples for qualification inspection, the manufacturer shall submit a request to the Aerospace Crew Equipment Department indicating a date on which samples can be forwarded and also request an authorization number to accompany the samples.

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6.3.1 When requested, the manufacturer shall submit engineering drawings and inspection reports in accordance with SD-6, Provisions Governing Qualification.

6.3.2 Qualified products list - For inclusion on a qualified products list, the PIRD and cartridge submitted for qualification shall successfully meet the test requirements of this specification and satisfy the following additional minimum requirements.

(a) Qualification and documentation of the cartridge shall be in accordance with paragraph 3.7.3 of MIL-D-21625.

(b) The cartridge actuated device shall be tested in accordance with paragraphs 3.11.1 and 3.11.2 of MIL-D-23615, except that six devices may be used in lieu of the twelve specified for the temperature and humidity test. Following the environmental treatments prescribed in MIL-D-23615, the device shall be actuated and shall meet the time and velocity requirements specified in 3.6.1.2. The -65° F, +70° F and +160° F firings of 4.7.7.1 shall be acceptable in lieu of the low, normal and high temperature tests of MIL-D-23615. For the purpose of actuation, a 98th percentile anthropomorphic dummy, dressed in full flight clothing and survival equipment in a manner acceptable to the testing activity, shall be installed in the seat with all personnel restraints connected. If the seat structure is not available then a suitable test fixture may be substituted upon approval by the cognizant Government activity.

6.4 Data - For the information of Contractors and Contracting Officers, any of the data specified in applicable documents listed in Section 2 of this specification, or referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The contractor shall be responsible for furnishing such design and test data required herein or by MIL-D-21625 and MIL-D-23615 for device qualification and release for service use. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order.

6.5 Explosives safety precautions -

6.5.1 Applicability - The loading, assembly and handling of the explosives, sub-assemblies and the finished items covered by this specification, involve hazardous operations and therefore require suitable explosives safety precautions. At plants of the Naval Air Systems Command, Department of the Navy, and other government plants where the government is responsible for safety, the precautions

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listed below are mandatory. These same minimum provisions listed below are recommended for observance by a contractor or manufacturer that explosively loads the item covered in this specification in order to fulfill his own safety responsibility. Other warnings and precautions, pertaining to the operational effectiveness or safety of the specified item during its use (for example, during its handling, launching or firing) are included in the detailed technical requirements of the specification. Use of this specification shall not be construed to replace other safety regulations of governmental agencies or to relieve the contractor or manufacturer of responsibility for the safety of his operations.

6.5.2 Personnel and equipment - Proper care must be exercised at all times to protect personnel from accidents, fires or explosions, and to limit damage to equipment and loading areas. In connection with the above, the precautionary measures in the following paragraphs should be observed. (See also 6.5.4, Static and Electrical and other applicable parts of these safety precautions.)

6.5.2.1 Explosives operations shall be kept under strict supervision. Periodic inspections shall be employed to assure compliance with established safety regulations.

6.5.2.2 Safe equipment and methods should be utilized for handling and transporting explosives and loaded parts.

6.5.2.3 Where required, remote control barricaded handling equipment shall be used for explosives operations such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

6.5.2.4 Employ properly proportioned and properly located protective barricades, screens, or shields at all required points.

6.5.2.5 Maintain equipment such as presses, dies, crimpers, blenders, mixers, sifters, conveyors, and the like, in safe operating condition.

6.5.2.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning, and associated industrial hazards.

6.5.2.7 During static tests, propulsion units such as rocket motors shall be restrained against full thrust to prevent flight if ignited, for example, during test of igniter circuit. Also, means to vent flames and exhaust gases harmlessly shall be provided.

6.5.3 Operating procedures - All loading operations shall be conducted in full accordance with a formalized operating procedure which shall include the following:

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6.5.3.1 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operations.

6.5.3.2 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.5.3.3 Appropriate safety instructions for specific operations shall be posted and personnel instructed as to the hazards present and precautions involved.

6.5.3.4 The number of working personnel and transients permitted in each work space shall be established and posted in such areas. The numbers shall be the minimum number required to accomplish necessary operations.

6.5.3.5 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings.

6.5.3.6 Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from, floors, walls, ceiling, ledges, tables, benches, piping, and equipment or the items loaded. Clean up any spilled material immediately.

6.5.3.7 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.5.3.8 Personnel handling detonators, primers, delay elements, leadins, boosters, and related parts which affect functioning should, insofar as practicable, avoid using bare fingers or improper equipment, in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.5.3.9 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges. (See 6.5.4.)

6.5.4 Static and electrical -

6.5.4.1 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges shall be employed. Fabrics, such as silk and nylon, which promote static generation shall be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted

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with items which are usually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low energy type electric primers, detonators and squibs. The latter types of items shall have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare non-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding from induced electric currents originated by sources such as lightning, static, electromagnetic radiations, etc. Where necessary for safety, humidity of work rooms should be appropriately increased, as required to lessen electro-static effects but without inducing excessive moisture absorption by any of the components of the item being processed.

6.5.4.2 Protect all explosive operations from effects of electric current originating in equipment such as soldering irons or guns, heaters, switches, wiring, motors, lights, test instruments, telephones, etc., by suitable insulation, grounding, separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or due to completing an electric circuit through an electric primer, detonator or squib. For example, circuits may be inadvertently completed from a defective electric soldering iron through a grounded contact. All electric type primers, detonators or squibs provided with wire leads shall have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Removable short circuiting clips, or other devices should be employed during manufacturing operations involving items such as electric primers, detonators or squibs.

6.5.4.3 Where necessary, protect personnel and operations from electromagnetic energy. In certain work with very sensitive explosives or items, a shielded environment free from significant electromagnetic energy should be provided.

6.5.5 Operating facility and storage regulations -

6.5.5.1 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance, at a minimum, with the American Table of Distances (ATD); and, while in process, in safety lockers and chests if in loading rooms, or in adequate ready or service magazines located in accordance with Intraplant Distances set forth in the ATD when outside of loading rooms. For Navy managed or contractor managed Navy owned explosives loading plants, the Armed Services Explosives Safety Board standards approved by the Department of Defense, the Secretary of the Navy and the Naval Air Systems Command, covering quantity-distance relations for explosives, apply.

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6.6 Changes from previous issue - Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-D-81514B(AS) Device Restraint Take-Up, Inertia-Locking Powered-Retracting; Gen

ORGANIZATION *(of submitter)* Spec for	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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